Introduction to Software Engineering

Project Idea & Team Contract



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1 Team Information

Item	Value
Team ID	04
Team name	TPNSM
Team lead	Nguyễn Võ Nhật Thuận
Communication channel	Messenger

Student ID	Full name	Email	Main role	Backup role
24127552	Nguyễn Võ Nhật Thuận	nvnthuan2410@clc.fitus.edu.vn	Team Lead	Business analyst / Full-stack developer
24127232	Lê Phụng Sơn	lpson2436@clc.fitus.edu.vn	Business analyst	Team Lead backup / Full-stack developer
24127087	Dương Nguyên Nam	dnnam2433@clc.fitus.edu.vn	Designer	Full-stack developer / Business Analyst
24127106	Lai Phạm Như Phúc	lpnphuc2431@clc.fitus.edu.vn	Full-stack developer	Tester
24127086	Trần Hoàng Khánh My	thkmy2432@clc.fitus.edu.vn	Tester	Designer / Full-stack developer

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

Project**:** SmartLib Booking - A smart web application for libraries that transforms manual room reservations into an automated, and an AI-assisted booking system.

2.2 Problem and Motivation

Currently, the demand for study rooms, group discussion spaces, and quiet study areas across the university library network is demanding. However, reservation management still heavily relies on a manual process: students submit booking requests via Google doc, and librarians manually update and track schedules using a shared Google Sheet.

2.3 Target Users

User group	What they need from the system
Student	Browse room availability, room types, and available equipment (e.g., projectors, whiteboards). Reserve study rooms instantly with confirmation using their student credentials. Manage personal bookings: view scheduled slots, receive automated reminder notifications, or cancel/reschedule bookings.
Librarians & Library Staff	Manage room inventories: add, update, or disable room profiles, update maintenance statuses, or adjust room capabilities. Replace manual Google Sheets tracking with automated approval rules and conflict detection.

2.4 Main Features

#	Feature	Short description	Must-have?
1	Basic Room Booking	Allows students to view available study rooms and submit reservation requests directly to the librarian for approval.	Yes
2	Room Inventory Management	Enables librarians to add/edit rooms with specific attributes (e.g., location/address, capacity, available amenities).	Yes
3	Automated First-Come-First-Served Mode	Provides a toggle switch for librarians to enable automatic booking approval, processing incoming student requests instantly without manual intervention on a first-come, first-served basis.	No
4	AI Assistant	Integrates an AI agent powered by a locally hosted Vision-Language Model (VLM) on the librarian's machine to handle natural language room searches, visual floor plan inquiries, and automated booking assistance.	No
5	Automated Status Notifications	Triggers automated email notifications when a booking is requested, approved, canceled.	No

2.5 Similar Products

- LibCal (by Springshare) :
    - Overview: The industry-standard web platform used by academic libraries worldwide for room, seat, and equipment reservations.
    - Limitations & Drawbacks:

        - High Complexity: The interface is overly enterprise-focused, bloated with features (event registration, billing, equipment checkouts) that make simple room bookings difficult.
        - Manual Staff Mediation: Booking requests still rely heavily on staff to review forms manually or use rigid static approval workflows.
- Skedda:
    - Overview: A modern SaaS workspace scheduling tool designed for corporate offices, coworking spaces, and higher-education institutions featuring interactive floor plans and automated rules.
    - Limitations & Drawbacks:
        - Commercial Focus: Optimized for paid desk/room rentals and corporate SSO setups rather than the specific non-commercial, academic workflows of university libraries.

Our Differentiating Solutions:

- VLM AI Assistant:
    - Unlike the systems requiring manual search filters, our system integrates a locally hosted Vision-Language Model (VLM) on the librarian's machine. This enables librarians and students to find rooms using natural language prompts without relying on expensive cloud AI services.
- Automated First-Come, First-Served Mode:
    - Eliminates the manual approval queue during peak study periods. Librarians can toggle an Auto-Approval Switch, allowing the system to process incoming student reservations instantly and conflict-free.
- User Experience:
    - Removes unnecessary enterprise bloat (billing, ticket sales) found in LibCal, focusing purely on rapid room discovery, flexible constraint management (group size, noise levels), and real-time automated status notifications via email.

2.6 Platform and Technology Ideas

- Platform: Web Application
- Frontend Framework: Django Templates + Tailwind CSS
- Backend Framework: Python
- Database & Infrastructure:
    - Database Engine: PostgreSQL.

    - Production / Cloud Database: Supabase Cloud.
    - Local Development Environment: Docker Container running local PostgreSQL instance.

2.7 Feasibility, Scope and Open Questions

- Feasibility :
    - The core Must-have features (Basic Room Booking, Room Inventory Management) native to Django capabilities. These can be reliably built, tested, and finalized within 2 weeks.
    - The Local VLM AI Assistant will be built using lightweight local vision-language API wrappers running on the librarian's machine.
- Out of Scope:
    - Online Payment / Fine Settlement System: No integration with real payment gateways for booking fees or late penalty charges.
    - Native Web Apps: Development is restricted to a Web Application without dedicated App Store or Play Store builds.
    - Multi-Library Enterprise Network: The system will focus on a single university library (e.g., VNU Central Library) rather than multi-tenant cross-campus routing.
- Main Risks:
    - Technical Risk - High Resource Consumption from Local VLM AI: *Risk:* Running a Vision-Language Model locally on a developer/librarian machine may cause high GPU/RAM consumption or slow response times during live demo grading.
- Open Questions for TA:

3. Team Contract

3.1 Goals and Expectations

- **Target Project Grade:** Achieve a score of 9.0 or higher.
- **Time Commitment:** At least 4 hours per week per member.

3.2 Communication

- **Main Communication Channel:** Messenger.
- **Message Response Time:** Within 12–24 hours on weekdays; within 12 hours close to deadlines.
- **Document & Task Repository:** GitHub Repository, Google Drive.

3.3 Meetings

- **Regular TA Meeting Schedule:** Thursday evenings, Online.
- **Internal Team Meeting Schedule:** Thursday noon, Online, weekly.
- **Absence Policy:** Members must provide at least 12 hours advance notice with a valid reason. Unexcused absences will be documented in the Weekly Status Report.
- **Internal Weekly Status Report Deadline:** Every Wednesday at 23:59.

3.4 Roles and Responsibilities

- Roles and responsibilities are specified in Section 1. Each member is primarily responsible for their main role and supports as a backup when necessary.
- The Team Lead or the backup Team Lead is responsible for compiling the Weekly Status Report and committing it to the repository on schedule.
- Attend all team meetings fully and stay focused during discussions.
- Provide advance notice with a valid reason if unable to attend a meeting.
- Acknowledge important announcement messages upon receiving them to confirm they have been read.

3.5 Work Standards

- **Internal Deadline:** All assigned deliverables and documentation sections must be completed at least 1 day before the official course submission deadline.
- **Quality Review:** Every document section and Pull Request (PR) must be reviewed and approved by at least one other team member before being finalized or merged into the main branch.
- **Task Management:** Every task must be created on the task board, self-assigned, and updated regularly with its completion status and due date.

3.6 Use of AI Tools

- **Allowed AI Tools:** ChatGPT, GitHub Copilot, Claude, Gemini, Codex.
- **Verification & Quality Policy:** All AI-generated outputs must be thoroughly verified, edited, and validated by team members for accuracy and suitability before being included in documentation or source code.

3.7 Decision Making

- **Decision-Making Method:** Decisions are made through discussion to reach consensus. If consensus cannot be reached, decisions will be determined by a majority vote.
- **Tie-Breaking Rule:** In the event of a tie vote, the Team Lead makes the final decision.

3.8 Handling Conflicts & Missed Work

- **Step 1:** Engage in a direct 1-on-1 discussion with the member involved to clarify the reasons and agree on a new completion deadline.
- **Step 2:** Discuss the issue openly in a team meeting and formally record the incident in the Weekly Status Report.
- **Step 3:** Report directly to the Teaching Assistant (TA). The severity of the violation will directly impact the individual contribution factor.

3.9 Signatures

By signing below, all team members commit to fully abiding by the rules and regulations outlined in this Team Contract.

Student ID	Full name	Signature	Date
24127552	Nguyễn Võ Nhật Thuận		11/10/2026
24127232	Lê Phụng Sơn		11/10/2026
24127087	Dương Nguyên Nam		11/10/2026
24127106	Lai Phạm Như Phúc		11/10/2026
24127086	Trần Hoàng Khánh My		11/10/2026

4. AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Võ Nhật Thuận	Gemini	Draft template for PA0 and translation of contract terms into English.	Reviewed with the team, filled in all project details, and customized rules to match requirements.